THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of coefficients in the balanced equation $\mathrm{Fe + H_2O \longrightarrow Fe_3O_4 + H_2}$ is: 1

(a) 9 **(b)** 10 **(c)** 12 **(d)** 8

2. In $\mathrm{Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe}$, the substance reduced is: 1

(a) Al **(b)** Fe **(c)** $\mathrm{Al_2O_3}$ **(d)** $\mathrm{Fe_2O_3}$

3. The colour of copper sulphate solution fades when an iron nail is dipped in it because: 1

(a) $\mathrm{FeSO_4}$ (pale green) is formed **(b)** water is formed

(c) iron turns blue **(d)** copper sulphate evaporates

4. The coefficient of $\mathrm{O_2}$ in the balanced equation $\mathrm{C_3H_8 + O_2 \longrightarrow CO_2 + H_2O}$ is: 1

(a) 5 **(b)** 7 **(c)** 3 **(d)** 4

5. In $\mathrm{ZnO + C \longrightarrow Zn + CO}$, the reducing agent is: 1

(a) C **(b)** CO **(c)** Zn **(d)** ZnO

6. Hydrogen and oxygen are obtained in the ratio 2 : 1 by volume because: 1

(a) oxygen is more soluble

(b) water contains 2 hydrogen atoms for each oxygen atom

(c) hydrogen is lighter

(d) oxygen is heavier

7. The coefficient of $\mathrm{O_2}$ in the balanced equation $\mathrm{KClO_3 \longrightarrow KCl + O_2}$ is: 1

(a) 1 **(b)** 4 **(c)** 3 **(d)** 2

8. $\mathrm{2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2}$ is a: 1

(a) precipitation **(b)** displacement

(c) photolysis **(d)** thermal decomposition

9. Chips packets are filled with nitrogen to: 1

(a) increase the weight

(b) keep them warm

(c) prevent oxidation of the oil in the chips

(d) make them react with air

10. The sum of coefficients in the balanced equation $\mathrm{Al + O_2 \longrightarrow Al_2O_3}$ is: 1

(a) 7 **(b)** 5 **(c)** 9 **(d)** 6

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between a physical change and a chemical change with one example each. 2

12. Balance $Al + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$ and name the gas evolved. 2

13. Why are silver bromide and silver chloride kept in dark bottles? 2

14. Why is the combustion of methane an exothermic reaction? Write the balanced equation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance $Al + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + H_2$ and state whether it is a displacement or a combination reaction. 3

16. Explain why the thermal decomposition of limestone needs continuous heating. Write the equation and one use of the solid product. 3

17. In the electrolysis of water, name the gas at each electrode, the ratio of volumes and the equation. 3

OR

Balance $NH_3 + O_2 \longrightarrow NO + H_2O$ and mention the type of reaction. 3

18. Zinc granules are added to blue copper sulphate solution. Give the observation, the equation and the reason. 3

OR

Lead nitrate is heated in a boiling tube. State three observations. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Classify each as combination, decomposition, displacement or double displacement: 5

(a) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$

(b) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$

(c) $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2\,HCl$

(d) $2\,Mg + O_2 \longrightarrow 2\,MgO$

(e) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$.

OR

Explain thermal, electrolytic and photolytic decomposition with one equation each. State which of the three is used in photography. 5

20. For 5

(i) $CuO + H_2 \longrightarrow Cu + H_2O$ and

(ii) $Fe_2O_3 + 3\,CO \longrightarrow 2\,Fe + 3\,CO_2$, identify the substance oxidised, reduced, and the oxidising and reducing agents.

OR

Explain in terms of oxygen how oxidation is involved in corrosion, rancidity, respiration and burning of fuel. 5

END OF QUESTION PAPER — **Total: 40 Marks**